SPATIAL AI + COMPUTING · COLUMBIA GSAPP

Spatial Agents

A delivery rider who learns Morningside Heights by riding it

Justin & Ishika

Simulation: ishika100603.github.io/Spatial_ai/food-delivery-3d.html



Kai’s first run

GOAL · Ana’s home, with Priya’s delivery on the way, then back to Wok on Wheels

START · knows only what it can see from the truck. No route is given

DEAD END · tries W 116th St: College Walk, no road

RE-PLAN · on its own map, by W 114th St

It also picks the order of the two deliveries itself.



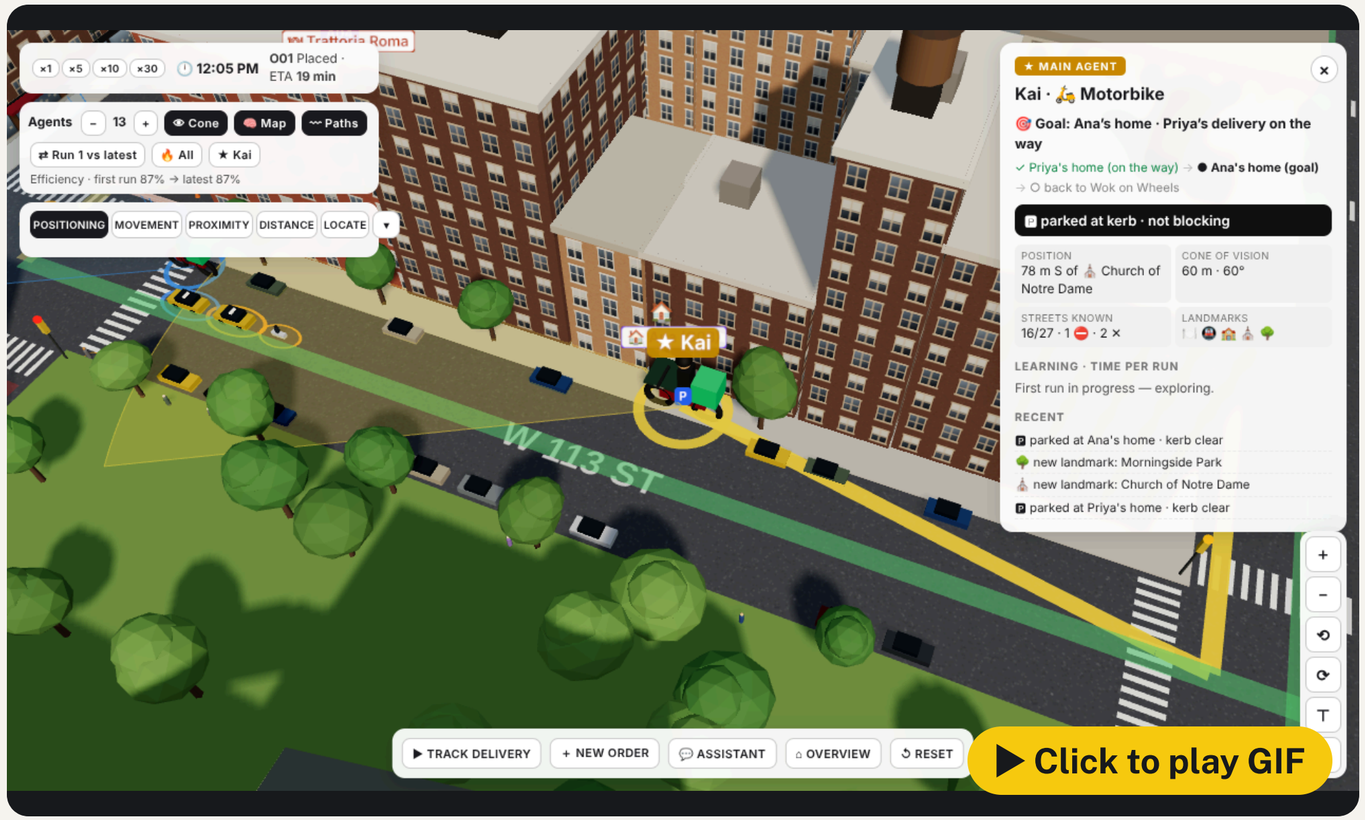
Cone of vision

MOTORBIKE · 60 m · 60°

CAR · 45 m · 90°

E-BIKE · 35 m · 120°

Buildings block it. Through it an agent finds closures, dead ends, one-way signs, landmarks and free kerb space.



What Kai knows

GREEN · streets it has seen

⛔ / ✕ · closures and streets with no road

P · kerb spots it parked in

LANDMARKS · restaurants, churches, schools, the park

Its position is told by its own landmarks: “78 m S of the Church of Notre Dame”.



Run 1 vs run 5

RUN 1 · 7.8 min · 1,686 m. Detours up Broadway to the campus streets

RUN 5 · 5.8 min · 1,527 m. Straight along W 114th St, no re-plans

It now skips the dead ends, finds doors without searching and parks in spots it remembers.



Every run, red to green

PATHS · all of Kai's runs, oldest red, newest green

CHART · minutes per run: 7.8, 7.0, 5.8, 7.0, 5.8

Run 4 took the same route as run 3; the extra time is traffic.



Among other agents

ON THE MOVE

Keeps a gap behind agents and cars in its lane

Waits for people crossing ahead

Stops at red lights

PARKING

At the kerb, at least 16 m from a crossing

Never in a doorway

Never against another parked agent

31 agents at once

STREET HEAT · the streets all agents use most, thick and orange

LANDMARK RINGS · how many agents know each landmark

Each courier carries two orders a run and learns on its own, like Kai.



Try it

ishika100603.github.io/Spatial_ai/food-delivery-3d.html

1. Press ★ **Kai** to follow the main agent, and set speed to ×30.
2. After a few runs, press ⇄ **Run 1 vs latest**.
3. Press **+** for more agents and 🔥 **All** for the shared view.

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